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Revision · Probability

Cambridge insert: putting a number on "how likely", and combining two events.

LESSON 99–100 | 2 × 40 MIN | ALGEBRA 8 — REVISION | STAGE 9 · 12.1–12.4

LESSON CLOCK

6' | 10' | 10' | 10' | 4'

Order five events by likelihood	6 min
The basic formula	10 min
Complement and mutually exclusive events	10 min
Two events	10 min
Homework	4 min

LEARNING OBJECTIVES

- Calculate the probability of an event from equally likely outcomes.
- Use the fact that probabilities of an event and its complement add to 1.
- Use a table or tree diagram for two events.
- Compare experimental with theoretical probability.

TEXTBOOK

National	Annual revision
Cambridge	Learner's Book pp. 255–278
Workbook	Workbook 12.1–12.4

01 Explanation

Putting a number on chance

FOR EQUALLY LIKELY OUTCOMES

$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

Every probability lies between 0 (impossible) and 1 (certain).

One die: $P(even) = \frac{3}{6} = \frac{1}{2}$, $P(7) = 0$, $P(less\ than\ 7) = 1$. Give the answer as a fraction in its simplest form unless a decimal or percentage is asked for.

The word "equally likely" is doing real work. A drawing pin can land point-up or point-down, but not with probability $\frac{1}{2}$ each — only an experiment can tell you.

Complement, and events that exclude each other

$$P(\text{not } A) = 1 - P(A)$$

If the probability of rain is 0.3, the probability of no rain is 0.7. This one line turns many "at least one" questions into a single subtraction — exactly as in the counting lesson.

MUTUALLY EXCLUSIVE EVENTS

If A and B cannot happen together, $P(A \text{ or } B) = P(A) + P(B)$.

A die shows a 2 or a 5: $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.

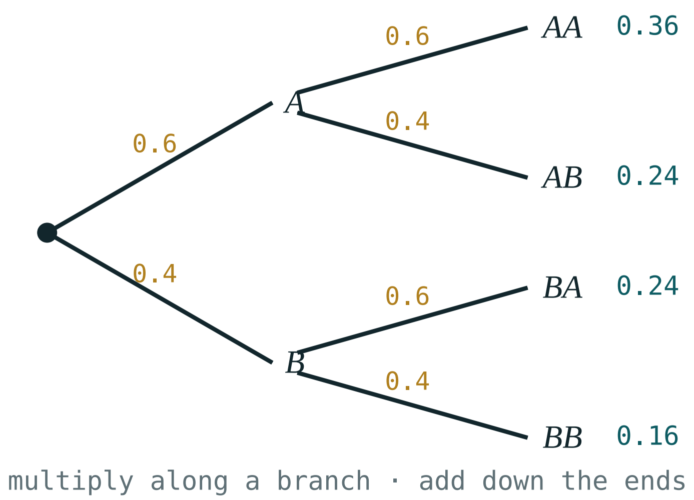
ONLY WHEN THEY EXCLUDE EACH OTHER

"Even" and "greater than 3" are not mutually exclusive — 4 and 6 belong to both — so adding the probabilities double-counts them.

Two events, and what experiments show

For two **independent** events, multiply along the branches of a tree:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$



Multiply along a branch to get the probability of that path; add the paths that satisfy the event.

Two coins: $P(\text{two heads}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. For "exactly one head" there are two successful paths, HT and TH, so add: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

	THEORETICAL	EXPERIMENTAL
Comes from	counting outcomes	actually doing the trials
Formula	$\frac{\text{favourable}}{\text{total}}$	$\frac{\text{times it happened}}{\text{times tried}}$
Behaviour	fixed	settles towards the theoretical value as trials increase

Expected frequency = probability × number of trials. In 300 throws of a die you expect about $300 \cdot \frac{1}{6} = 50$ sixes — “about”, never exactly.

02 Worked examples

EXAMPLE 1

A bag holds 5 red, 3 blue and 4 green counters. One is taken at random. Find $P(red)$ and $P(not\ green)$.

1. $5 + 3 + 4 = 12$

   Total outcomes.

2. $P(red) = \frac{5}{12}$

3. $P(green) = \frac{4}{12} = \frac{1}{3}$

4. $P(not\ green) = 1 - \frac{1}{3} = \frac{2}{3}$

   Use the complement.

ANSWER

$P(red) = \frac{5}{12}$, $P(not\ green) = \frac{2}{3}$

EXAMPLE 2 **Two dice are thrown. Find the probability that the total is 9.**

1. 36 equally likely outcomes

   A 6 by 6 table.

2. $(3,6), (4,5), (5,4), (6,3)$

   Four ways to make 9.

3. $P = \frac{4}{36} = \frac{1}{9}$

ANSWER

$\frac{1}{9}$

EXAMPLE 3 **The probability that a bus is late is 0.2. Find the probability that it is late on both of two days, and that it is late on exactly one.**

1. $P(late, late) = 0.2 \cdot 0.2 = 0.04$

   Independent days — multiply.

2. $P(late, on\ time) = 0.2 \cdot 0.8 = 0.16$

   First branch.

3. $P(on\ time, late) = 0.8 \cdot 0.2 = 0.16$

   Second branch.

4. $0.16 + 0.16 = 0.32$

   Add the two successful paths.

ANSWER

0.04 and 0.32

03 Interactive model

Work through one probability question at a time; ask for the answer before revealing each step.

Probability, step by step

> A card numbered 1 to 20 is drawn. Find $P(\text{multiple of } 3)$.

1. $3, 6, 9, 12, 15, 18$

   List the favourable outcomes.

2. 6 favourable out of 20

3. $\frac{6}{20} = \frac{3}{10}$

   Simplify.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Probability	Ehtimollik	Вероятность
Event	Hodisa	Событие
Equally likely	Teng imkoniyatli	Равновозможные
Favourable outcome	Qulay natija	Благоприятный исход
Complement	Qarama-qarshi hodisa	Противоположное событие
Mutually exclusive	Birgalikda bo‘lmaydigan	Несовместные
Independent events	Bog‘liq bo‘lmagan hodisalar	Независимые события
Tree diagram	Daraxt diagrammasi	Дерево вероятностей
Experimental probability	Tajribaviy ehtimollik	Опытная (частотная) вероятность
Expected frequency	Kutilayotgan chastota	Ожидаемая частота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A die is thrown. $P(\text{more than } 4)$ is:

- $\frac{1}{6}$
- $\frac{1}{3}$
- $\frac{1}{2}$
- $\frac{2}{3}$

If $P(A) = 0.42$ then $P(\text{not } A)$ is:

- 0.42
- 0.58
- 0.68
- 1.42

Two coins are tossed. $P(\text{exactly one head})$ is:

- $\frac{1}{4}$
- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{3}{4}$

A spinner has $P(\text{blue}) = 0.25$. In 80 spins expect about:

- 16
- 20
- 25
- 40

Which pair is NOT mutually exclusive for one die?

- “2” and “5”
- “even” and “odd”
- “even” and “more than 3”
- “1” and “6”

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *A die is thrown. Find P(3).*

   ANSWER $\frac{1}{6}$

2. *A die is thrown. Find P(odd).*

   ANSWER $\frac{1}{2}$

3. *A bag has 3 red and 7 blue. Find P(red).*

   ANSWER $\frac{3}{10}$

4. *If P(win) = 0.4, find P(not win).*

   ANSWER *0.6*

5. *A coin is tossed twice. Find P(two tails).*

   ANSWER $\frac{1}{4}$

6. *A card is drawn from 1–10. Find P(more than 7).*

   ANSWER $\frac{3}{10}$

7. *A spinner has P(red) = 0.5. In 60 spins, expect how many reds?*

   ANSWER *30*

Medium · the standard question

7 PROBLEMS

1. *Two dice are thrown. Find P(total 7).*

ANSWER $\frac{1}{6}$

2. *A bag has 4 red, 5 blue, 6 green. Find P(not blue).*

ANSWER $\frac{2}{3}$

3. *P(late) = 0.15. Find P(late on both of two days).*

ANSWER 0.0225

4. *A coin is tossed three times. Find P(exactly two heads).*

ANSWER $\frac{3}{8}$

5. *A card is drawn from 1–20. Find P(prime).*

ANSWER $\frac{8}{20} = \frac{2}{5}$

6. *P(A) = 0.3, P(B) = 0.5 and A, B are mutually exclusive. Find P(A or B).*

ANSWER 0.8

7. *A spinner has P(green) = 0.35. In 400 spins, expect how many greens?*

ANSWER 140

Hard · stretch and reason

7 PROBLEMS

1. *A bag has 5 red and 4 blue. Two are taken without replacement. Find P(both red).*

   ANSWER $\frac{5}{9} \cdot \frac{4}{8} = \frac{5}{18}$

2. *Two dice are thrown. Find P(at least one 6).*

   ANSWER $\frac{11}{36}$

3. *P(rain) = 0.4 on each of three days. Find P(no rain at all).*

   ANSWER $0.6^3 = 0.216$

4. *P(rain) = 0.4 on each of three days. Find P(at least one wet day).*

   ANSWER $1 - 0.216 = 0.784$

5. *A bag has 6 red and 4 blue. One is taken and replaced, then another. Find P(different colours).*

   ANSWER $2 \cdot 0.6 \cdot 0.4 = 0.48$

6. *A die is thrown 120 times and gives 32 sixes. Compare experimental and theoretical probability.*

   ANSWER $\frac{32}{120} \approx 0.27$ against $\frac{1}{6} \approx 0.17$ — noticeably high; either chance, or the die is biased

7. *Explain why P(A or B) = P(A) + P(B) can fail.*

   ANSWER If A and B can happen together, the outcomes in both are counted twice; the correct rule subtracts *P(A and B)*.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 12. Simplify every fraction.

1. *A bag has 7 red and 5 blue. Find P(blue) and P(not blue).*
2. *Two dice are thrown. Find P(total 5).*

3. *A coin is tossed three times. Find P(at least one tail).*
4. *P(faulty) = 0.05. In 600 items, how many faulty ones are expected?*
5. *A card is drawn from 1–15. Find P(multiple of 4).*
6. Explain the difference between experimental and theoretical probability in two sentences.